

# Ojas Pradhan

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## EDUCATION

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### Drexel University

Ph.D., Architectural Engineering  
M.Sc., Mechanical Engineering  
B.Sc., Mechanical Engineering

Philadelphia, PA  
Expected – June 2023  
June 2019  
June 2019

## RESEARCH INTERESTS

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Data-driven fault investigation, use of artificial intelligence for sustainable built environment, smart building control

## RESEARCH EXPERIENCE

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### Graduate Research Assistant

*Drexel University*

June 2019 - Present

*Philadelphia, PA*

- Contribute to multi-disciplinary projects aimed at developing automated data-driven strategies to detect, diagnose and predict HVAC faults in buildings
- Develop and deploy data predictive supervisory control to minimize energy consumption for multi-stream building systems
- Collaborate and coordinate with faculty, staff scientists, and fellow graduate students across various universities and research laboratories

### Undergraduate Research Assistant

*Drexel University*

October 2017 - June 2019

*Philadelphia, PA*

- Implemented various whole building HVAC system faults in difference campus buildings to test fault diagnosis algorithms
- Analyzed daily operational data for chiller and air handler units located at demo buildings
- Evaluated fault impact on energy consumption and system operation
- Performed literature review and comparison studies to evaluate existing dynamic building simulation environments
- Developed user and data interface to simulate dynamic hydronic system of a research laboratory using TRNSYS and MATLAB
- Recorded and published simulation results for future deployment of intelligent control algorithms to improve building system operation

## TECHNICAL SKILLS

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**Programming Languages:** Python, Matlab, R, SAS, HTML/CSS

**Data Science:** Machine Learning (Keras/Tensorflow, PyTorch), Deep Learning (CNNs, RNNs), Statistical Inferences, Data Visualization

**Database Management:** SQL (Postgres, MariaDB), NoSQL (MongoDB)

**Cloud-based Technologies:** Microsoft Azure, Docker

**Engineering Tools:** Dymola, Simulink, TRNSYS, EnergyPlus, LabVIEW

**Others:** Git, LaTeX

## JOURNAL PUBLICATIONS

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1. Huang, J., Yoon, H., Wu, T., Candan, K. S., **Pradhan, O.**, Wen, J., & O'Neill, Z. (2022). Eigen-Entropy: A Metric on Multi-variate Sampling Decision. Information Sciences. (Accepted)
2. Huang, J., Yoon, H., **Pradhan, O.**, Wu, T., Wen, J., O'Neill, Z., & Candan, K. S. (2022). A Cosine-based Correlation Information Entropy Approach for Building Automatic Fault Detection Baseline Construction. Science and Technology for the Built Environment.

3. Huang, J., Wen, J., Yoon, H., **Pradhan, O.**, Wu, T., O'Neill, Z., & Candan, K. S. (2022). Real vs. Simulated: questions on the capability of simulated datasets on building fault detection for energy efficiency from a data-driven perspective. *Energy and Buildings*, 111872
4. Chen, Y., Wen, J., **Pradhan, O.**, Lo, J., & Wu, T. (2021). Using Discrete Bayesian Networks for Diagnosing and Isolating Cross-level Faults in HVAC Systems. *Applied Energy* (Under Review)

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## CONFERENCE PRESENTATIONS

1. **Pradhan, O.**, Hälleberg, D., Chen, Z., Wen, J., Wu, T., Candan, K. S., & O'Neill, Z (2022). Lagged-kNN Based Data Imputation Approach for Multi-Stream Building Systems Data. 7th International High-Performance Buildings Conference at Purdue. July 11-14, 2022.
2. **Pradhan, O.**, Wen, J., Chen, Y., Lu, X., Chu, M., Fu, Y., O'Neill, Z, Wu, T., & Candan, K. S. (2021). Dynamic Bayesian network-based fault diagnosis for ASHRAE guideline 36: high performance sequence of operation for HVAC systems. In *Proceedings of the 8th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation* (pp. 365-368)
3. Huang, J., Wu, T., Yoon, H., **Pradhan, O.**, Wen, J., & O'Neill, Z. (2021). Automatic Fault Detection Baseline Construction for Building HVAC Systems using Joint Entropy and Enthalpy. In *IIE Annual Conference Proceedings* (pp. 536- 541)
4. **Pradhan, O.**, Wen, J., Chen, Y., Wu, T., & O'Neill, Z. (2021). Dynamic Bayesian Network for Fault Diagnosis. 2021 ASHRAE Virtual Conference. June 28 – June 30, 2021
5. **Pradhan, O.**, Pertzborn, A., Zhang, L., & Wen, J. (2020). Development and Validation of a Simulation Testbed for the Intelligent Building Agents Laboratory (IBAL) using TRNSYS. 2020 ASHRAE Annual Conference. June 29 – July 1, 2020
6. Chen, Y., Wen, J., Chen, T., & **Pradhan, O.** (2018). Bayesian Networks for Whole Building Level Fault Diagnosis and Isolation. 5th International High-Performance Buildings Conference at Purdue. July 9-12, 2018

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## PROFESSIONAL EXPERIENCE

### Mechanical Engineering Coop

*Bristol-Myers Squibb Company*

April – September 2017

*Hopewell, NJ*

- Designed, programed and troubleshooted industrial control systems, such as motion/servo controllers, VFDs, PLCs, HMIs along with SCADA and SQL databases
- Researched on energy markets, plant thermal efficiencies, utility rate structures, commodity pricing and regulatory requirements
- Developed engineering solutions and managed projects involving retrofits, optimization and acquisition of process systems
- Verified project benefits and energy savings through pre/post energy audits
- Maintained and documented equipment/system modifications for all cGMP, vivarium and lab spaces
- Utilized Autodesk application suite to review, update and file drawings, records and documentations for new turn over packages

### Engineering Intern

*Philadelphia Water Department*

April 2016 – September 2016

*Philadelphia, PA*

- Created standard procedures, references, templates and other resources to streamline the planning and design processes
- Tracked project and program metrics; performed cost analyses and reported program data on company database
- Conducted research studies on existing design practices and provided recommendations for future design specifications
- Collaborated with other units within the department such as Design, Monitoring and Maintenance to develop and promote green stormwater infrastructure throughout Philadelphia

## HONORS AND AWARDS

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- Linda Latham Scholarship - Presented by American Council for an Energy-Efficient Economy (2022)
- Invited PhD Fellow at University of Nebraska-Lincoln (2022)
- ASHRAE Graduate Student Scholarship – Presented by ASHRAE Philadelphia Chapter (2020)
- ASHRAE Undergraduate Student Scholarship – Presented by ASHRAE Philadelphia Chapter (2019)
- Sustainable Energy Fund Scholarship – Presented by Sustainable Energy Fund (2018)
- Best Peer Mentor – Presented by Academic Center for Engineers, Drexel University (2018)
- Himalaya Youth Forum Invitee – Presented by ICIMOD (2018)

## PROFESSIONAL ASSOCIATIONS

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- American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE), Student Member
- Association for Computing Machinery (ACM), Member
- Energy in Buildings and Communities (EBC): Annex 81 – Data Driven Smart Buildings

## REFERENCES

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### **Jin Wen, PhD.**

Professor

Department of Civil, Architectural and Environmental Engineering

Drexel University

jinwen@drexel.edu

### **Yimin Chen, PhD.**

Senior Scientific Engineering Associate

Building Technology & Urban Systems Division

Lawrence Berkeley National Laboratory

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